

FIBER OPTIC SPLICE ENGINEERING SHEET

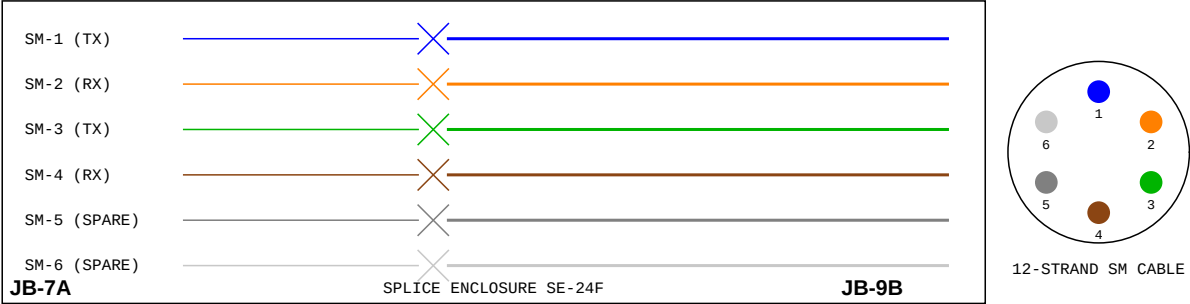
Zava Telecom -- Network Engineering Division

DOC-FSE-2026-04-28-0041

DOCUMENT INFORMATION

Field	Value	Field	Value
Doc ID	FSE-2026-0041	Revision	C (2026-04-28)
Project	Tower Ridge Rehab	Engineer	K. Nakamura, PE
Site	Site B -- Segment 9	Approved	M. Webb (Ops Mgr)
Cable Type	12-SM OS2 Tight	Status	AS-BUILT

FIBER ROUTING DIAGRAM -- SPLICE DETAIL



STRAND TEST RESULTS -- POST-SPLICE VERIFICATION

Strand	Color	Type	Loss(dB/km)	Status	OTDR Ref
SM-1	Blue	G.652D	0.19	PASS	OTDR-0041-01
SM-2	Orange	G.652D	0.21	PASS	OTDR-0041-02
SM-3	Green	G.652D	0.18	PASS	OTDR-0041-03
SM-4	Brown	G.652D	0.22	PASS	OTDR-0041-04
SM-5	Slate	G.652D	0.52*	MARGINAL	OTDR-0041-05
SM-6	White	G.652D	0.17	PASS	OTDR-0041-06
SM-7	Red	G.652D	0.48*	MARGINAL	OTDR-0041-07
SM-8	Black	G.652D	0.20	PASS	OTDR-0041-08
SM-9	Yellow	G.652D	0.19	PASS	OTDR-0041-09
SM-10	Violet	G.652D	0.21	PASS	OTDR-0041-10
SM-11	Rose	G.652D	0.55*	FAIL	OTDR-0041-11
SM-12	Aqua	G.652D	0.18	PASS	OTDR-0041-12

* Elevated attenuation. Strands 5, 7, 11 showing stress-induced loss. See condition report CR-2026-04-10.

FIELD NOTE (P. Sharma, 04/28):
Strand 11 re-spliced but loss unchanged.
Micro-bend at TV-3 (confirmed VID-05, AUD-04-15).
DO NOT re-splice. Conduit is root cause.
Full conduit replacement required first.

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K. Nakamura, PE  
Senior Fiber Engineer  
Date: 2026-04-28

### --- ENGINEERING ANALYSIS (v3) ---

Loss trend (SM-11): 0.42 dB/km (Jan) -> 0.47 (Feb) -> 0.55 (Apr 28)  
Progressive degradation confirms mechanical cause (not fusion quality).

CRITICAL FINDING: Route 2 shares conduit with Route 1 in Sections 3-5.  
Total customer exposure: 34 (Rt1) + 8 (Rt2) = 42 with NO redundancy.  
Conduit installed 2019 (pre-ZNS-2021-RED-004 redundancy standard).

RECOMMENDATION: Aerial bypass for Route 2 BEFORE conduit replacement.  
See FRD-2026-04-28 for routing analysis and reroute options.  
Materials ordered: PO-2026-0391 (Webb, 05/03). Budget: BUD-2026-TR-009.